USN | | | | | | | | | | [23CE33]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

MECHANICS OF MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
			BL	**CO**	**M**
1.	**a.**	Define stress and strain.	1	1	**02**
	b.	What is Poisson's ratio?	1	1	**02**
	c.	Define neutral axis in bending.	1	2	**02**
	d.	What is the difference between shear force and bending moment?	1	3	**02**
	e.	Define polar moment of inertia.	1	4	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Explain Hooke's law and give the relationship among elastic constants.	3	1	**05**
		OR			
	b.	Derive the relationship between Young's modulus, Bulk modulus, and Poisson's ratio.	3	1	**05**
3.	**a.**	Derive the bending moment at the midpoint of a simply supported beam subjected to a concentrated load 'W' kN at the center.	3	2	**05**
		OR			
	b.	Derive the shear force and bending moment diagram for a cantilever beam subjected to a uniformly distributed load 'W' kN	3	2	**05**
4.	**a.**	Derive the equation for the bending stress in a beam using the bending equation M/I = f/y = E/R	3	3	**05**
		OR			
	b.	Determine the section modulus for a circular section of diameter 100 mm.	3	3	**05**
5.	**a.**	Compare the weights of a solid and hollow shaft for the same strength.	3	4	**05**
		OR			
	b.	Calculate the diameter of a shaft required to transmit 150 HP at 1200 RPM, with a maximum shear stress of 60 MPa.	3	4	**05**

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6.	**a.**	Determine the principal stresses in a 2D stress system where σ_x = 80 MPa, σ_y = 30 MPa, and τ_{xy} = 20 MPa.	3	5	**05**
		OR			
	b.	Derive the formula for circumferential stress in a thin cylindrical shell subjected to internal pressure.	3	5	**05**
7.	**a.**	Define torsion and derive the torsion equation.	3	4	**05**
		OR			
	b.	Calculate the angle of twist for a solid shaft of length 3 meters, diameter 50 mm, and subjected to a torque of 1000 Nm. Take G = 80 GPa.	3	4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Calculate the stress in a bar of varying section subjected to an axial load.	3	1	**10**
		OR			
	b.	A bar of 1.5 m length is fixed at both ends and is subjected to a temperature increase of 50°C. Determine the stress developed if the coefficient of thermal expansion is 12 x 10^(-6)/°C, and Young's modulus is 200 GPa.	4	1	**10**
9.	**a.**	Draw and explain the bending moment and shear force diagrams for a simply supported beam with an overhang subjected to uniformly distributed load.	3	2	**10**
		OR			
	b.	A beam is subjected to three-point loads and a uniformly distributed load. Draw the shear force and bending moment diagram and determine the maximum BM and SF.	3	2	**10**
10.	**a.**	A rectangular beam has a width of 200 mm and depth of 400 mm. It is subjected to a bending moment of 50 kN-m. Calculate the maximum bending stress.	4	3	**10**
		OR			
	b.	Calculate the section modulus for an I-section with flange width = 150 mm, depth = 300 mm, and web thickness = 10 mm.	4	3	**10**
11.	**a.**	A hollow shaft has an external diameter of 150 mm and an internal diameter of 100 mm. Calculate the maximum shear stress if the shaft is subjected to a torque of 10 kN-m.	4	4	**10**
		OR			

	b.	Compare the weight and strength of a hollow shaft with external diameter 120 mm and internal diameter 60 mm with a solid shaft of 100 mm diameter for the same material.	4	4	**10**
12.	**a.**	Determine the principal planes and principal stresses in a member subjected to stresses σ_x = 100 MPa, σ_y = 50 MPa, and τ_{xy} = 25 MPa. Use Mohr's circle.	4	5	**10**
		OR			
	b.	Derive the formula for longitudinal and circumferential stresses in a thin cylindrical shell subjected to internal pressure.	3	5	**10**
13.	**a.**	A cylindrical shell has a diameter of 1 m, thickness of 10 mm, and is subjected to an internal pressure of 2 MPa. Calculate the change in diameter and volume.	3	4	**10**
		OR			
	b.	A cylindrical shell is subjected to internal pressure. Derive the expressions for longitudinal and volumetric strains.	3	4	**10**